

# Knowledge Representation & Planning

## Planning in Artificial Intelligence

### 1. Introduction to Planning in AI

Planning is the process of generating a sequence of actions that transforms an initial state into a goal state.

In AI:

- The agent must decide what actions to perform.
- Actions change the environment.
- The goal is to reach a desired state.

Planning is widely used in:

- Robotics
- Automated scheduling
- Game AI
- Intelligent agents
- Logistics systems

### 2. Components of a Planning Problem

A planning problem consists of:

1. Initial State  
Description of the starting situation.
2. Goal State  
Desired condition to be achieved.
3. Actions (Operators)  
Each action has:
  - Preconditions (what must be true before action)
  - Effects (how the world changes after action)
4. State Space  
All possible configurations of the world.

### 3. Basic Representation of Plans

A plan is a sequence of actions that achieves the goal from the initial state.

General structure:

Initial State → Action1 → Action2 → ... → Goal State

#### 3.1 Action Representation (STRIPS(Stanford Research Institute Problem Solver) Representation)

In classical planning, actions are represented using STRIPS format.

Each action contains:

1. Preconditions
2. Add list (positive effects)
3. Delete list (negative effects)

Example:

Action: Pickup(A)

Preconditions:

- OnTable(A)
- Clear(A)
- HandEmpty

Add list:

- Holding(A)

Delete list:

- OnTable(A)
- Clear(A)
- HandEmpty

#### 3.2 Example of a Simple Plan

Initial State:

- At(Home)
- HaveMoney

Goal:

- At(Office)

Actions:

1. Drive(Home, Office)
  - Precondition: At(Home)
  - Effect: At(Office)

Plan:

Drive(Home, Office)

## 4. Partial Order Planning (POP)

Partial Order Planning is a planning technique where actions are not totally ordered.

Instead of fixing a strict sequence, only necessary ordering constraints are specified.

Also called:

Non-linear Planning

### 4.1 Why Partial Order Planning?

In many problems:

- Some actions are independent.
- They do not need strict ordering.
- Forcing full sequence reduces flexibility.

POP allows:

- Flexible execution
- Parallel actions
- Efficient search

### 4.2 Key Concepts in Partial Order Planning

1. Steps  
Actions in the plan.
2. Ordering Constraints  
Specifies which action must occur before another.
3. Causal Links  
A link between actions where:  
Action1 produces a condition needed by Action2.
4. Open Preconditions  
Preconditions that are not yet satisfied.

## 4.3 Example of Partial Order Planning

Goal:

HaveTea and ReadBook

Actions:

1. BoilWater
2. MakeTea
3. PickBook
4. ReadBook

Constraints:

BoilWater → MakeTea

PickBook → ReadBook

There is no need to order:

MakeTea and PickBook

They can occur independently.

This is partial ordering.

## 4.4 Advantages of Partial Order Planning

- Avoids unnecessary ordering
- Supports parallel execution
- More efficient than linear planning
- Flexible solution representation

## 5. Planning in the Blocks World

### नया उदाहरण : Blocks World में Planning

#### प्रारंभिक स्थिति (Initial State)

On(A, Table)  
On(B, A)  
On(C, Table)  
Clear(B)  
Clear(C)  
HandEmpty

स्थिति का अर्थ:

B, A के ऊपर रखा है।  
A टेबल पर है।  
C टेबल पर है।  
B और C के ऊपर कोई ब्लॉक नहीं है।  
हाथ खाली है।

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#### लक्ष्य (Goal State)

On(A, C)  
On(B, A)

मतलब:

A को C के ऊपर रखना है।  
B को A के ऊपर रखना है।

अंतिम संरचना इस प्रकार होगी:

B  
A  
C

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#### समस्या को समझना

अभी स्थिति है:

B  
A C

लेकिन लक्ष्य के अनुसार पहले A को C पर रखना है।  
समस्या यह है कि A के ऊपर B रखा हुआ है।  
इसलिए पहले B को हटाना आवश्यक है।

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## समाधान की योजना (Planning Steps)

Unstack(B, A)

क्योंकि B, A के ऊपर है, B clear है और हाथ खाली है।  
अब B हाथ में आ जाता है।

Putdown(B)

B को टेबल पर रख देते हैं।  
अब A clear हो जाता है और हाथ खाली हो जाता है।

Pickup(A)

A टेबल पर है और clear है।  
अब A हाथ में है।

Stack(A, C)

C clear है और हाथ में A है।  
अब A, C के ऊपर रख दिया जाता है।

Pickup(B)

B टेबल पर है और clear है।  
अब B हाथ में है।

Stack(B, A)

A clear है और हाथ में B है।  
अब B को A के ऊपर रख दिया जाता है।

अंतिम स्थिति:

On(B, A)

On(A, C)

लक्ष्य प्राप्त हो गया।

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## इस उदाहरण से क्या सीखते हैं?

कभी-कभी लक्ष्य तक पहुँचने के लिए पहले बाधा को हटाना पड़ता है।

Subgoal interaction होता है — एक लक्ष्य पूरा करने से पहले दूसरे हिस्से को संभालना पड़ता है।  
सही क्रम में क्रियाओं की योजना बनाना ही Planning कहलाता है।

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## परीक्षा के लिए छोटा उत्तर

ब्लॉक्स वर्ल्ड में प्रारंभिक स्थिति से लक्ष्य स्थिति तक पहुँचने के लिए उपयुक्त क्रियाओं का क्रम बनाना प्लानिंग कहलाता है। इस उदाहरण में पहले B को हटाया गया, फिर A को C पर रखा गया और अंत में B को A पर रखकर लक्ष्य प्राप्त किया गया।

## 6. Challenges in Blocks World Planning

1. Subgoal interaction  
Achieving one goal may disturb another.
2. Search complexity  
State space grows rapidly.
3. Dead-end states

Blocks World is used to:

- Study heuristic search
- Study planning algorithms
- Test AI planners

## 7. Comparison: Linear vs Partial Order Planning

Linear Planning:

- Actions executed in strict sequence
- Simpler but less flexible

Partial Order Planning:

- Only necessary order is defined
- More flexible
- Allows parallel execution

## 8. Applications of Planning in AI

1. Robotics
2. Automated manufacturing
3. Game AI
4. Intelligent scheduling
5. Space mission planning
6. Autonomous vehicles

## 9. Exam-Oriented Short Definitions

Planning:

Process of generating a sequence of actions to achieve a goal.

Partial Order Planning:

A non-linear planning technique where actions are partially ordered based on necessity.

Blocks World:

A classical AI problem used to study planning and reasoning using blocks stacked on each other.

# Planning in Agentic AI, LLMs and GenAI (Short Notes)

## 1. Planning in Agentic AI

Agentic AI systems:

- Set goals
- Break goals into sub-goals
- Execute actions
- Observe results and re-plan

This is similar to classical AI planning (Initial State → Actions → Goal State).

Example:

Goal: "Create a research report"

Agent plan:

1. Search data
2. Summarize
3. Generate report
4. Format document

This follows goal decomposition and sequential planning.

## 2. Planning in LLMs

LLMs do not use explicit symbolic planners.

They perform implicit planning learned from training data.

Example:

Prompt: "How to start a business?"

LLM generates steps:

1. Market research
2. Register company
3. Arrange funding

This resembles forward reasoning.

## 3. Planning in Generative AI

GenAI performs planning when generating structured outputs.

Example:

Prompt: "Create a website"

Model internally plans:

1. HTML structure
2. CSS styling
3. Add content

Even though planning is not explicit, output follows logical step order.



## 4. Key Difference

Classical Planning:

- Symbolic
- Rule-based
- Explicit states

LLMs and GenAI:

- Learned patterns
- Probabilistic reasoning
- Implicit planning

Agentic AI combines both:

LLM reasoning + structured planning + tool execution.